



FAST – National University of Computer and Emerging Sciences

Department of Computer Science

StayBuddy

Smart Hostel Recommendation System

Final Year Project Report

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Chapter 1

Introduction

Finding suitable accommodation is a common challenge for many users, especially those who relocate to a new city for education or work. Users often rely on informal sources such as social media groups, online advertisements, or word of mouth to locate hostels. These methods frequently provide incomplete or unreliable information regarding hostel facilities, safety, cleanliness, and availability.

StayBuddy is proposed as a smart hostel discovery and recommendation platform designed to help users easily find suitable accommodation. The system provides verified hostel listings, user reviews, booking functionality, and personalised recommendations based on user interactions. The main goal of StayBuddy is to simplify the hostel search process and provide users with reliable and data-driven suggestions when selecting accommodation.

1.1 Background

The rapid growth of higher education in Pakistan has led to a surge in student migration to major cities like Islamabad, Lahore, and Karachi. According to the Higher Education Commission (HEC), over 1.6 million students are enrolled in universities across the country, with a significant percentage requiring off-campus accommodation. Existing solutions such as OLX, Zameen.com, and Facebook groups are fragmented, lack verification, and offer no intelligence. This gap motivated the development of StayBuddy – a centralised, AI-powered platform that streamlines hostel discovery, booking, and management.

1.2 Existing Solutions

Several accommodation platforms exist such as Airbnb and Booking.com that allow users to search and book accommodation. These platforms provide detailed listings, ratings, and reviews from previous users. However, they mainly focus on short-term stays and tourism rather than long-term hostel accommodation. In many regions, hostel searching still depends on unstructured platforms such as Facebook groups, classified advertisements, or personal referrals.

Research in recommendation systems shows that analysing user behaviour such as views, ratings, and bookings can help recommend suitable options. Collaborative filtering and hybrid

recommendation models are widely used in modern recommendation platforms. StayBuddy aims to apply similar techniques to hostel discovery by analysing user interactions and reviews to generate intelligent hostel recommendations.

Table 1.1: *Comparison of Existing Solutions*

System Name	System Overview	System Limitations
Airbnb / Booking.com	Global short-term rental platform with reviews and maps.	Focus on tourism; not optimised for student hostels or long-term stays; no parent/warden features.
OLX / Zameen.com	Classified ads for property rental.	Unverified listings; no recommendation engine; no booking or complaint management.
Facebook Groups	Informal community pages where hosts post vacancies.	Unstructured; high spam; no trust or personalisation; no AI support.

1.3 Problem Statement

Users currently face several difficulties when searching for hostels:

- Lack of a centralised platform containing verified hostel listings.
- Limited access to trustworthy reviews about hostel facilities.
- Difficulty comparing hostels based on price, cleanliness, and facilities.
- Lack of personalised recommendations based on user preferences.
- Parents have no visibility into their child's accommodation status.
- Wardens lack efficient tools to manage complaints and attendance.

Because of these limitations, users often choose accommodation that does not match their needs or expectations. The proposed StayBuddy system addresses these issues by providing a centralised platform where users can search hostels, read reviews, compare facilities, receive personalised recommendations, and manage bookings with full stakeholder involvement (students, parents, wardens, owners).

1.4 Scope

The scope of the StayBuddy system includes:

- Hostel discovery and listing management.
- User interaction tracking.
- Hostel reviews and ratings.
- Booking management with AI-assisted room selection.
- Complaint management with AI-suggested resolutions.
- Recommendation system for personalised hostel suggestions.
- Parent portal for monitoring child's attendance and bookings.
- Warden dashboard for managing complaints, attendance, and announcements.
- Owner module for adding and managing hostels.
- Admin portal for verification and user management.

The system is developed as a web-based application consisting of a React frontend, FastAPI backend, PostgreSQL database, and Python-based AI microservices for recommendations and natural language processing.

1.5 Modules

1.5.1 User Management Module

Handles user registration, login, profile management, and role-based access control (Student, Parent, Warden, Owner, Admin).

- JWT-based authentication with role-specific dashboards.
- Student preference management (budget, amenities, commute distance).

1.5.2 Hostel Management Module

Stores hostel information including location, capacity, facilities, room types, pricing, and verification status.

- Owner adds/edits hostel details.

- Admin verifies hostels before public listing.

1.5.3 Booking Management Module

Allows users to reserve rooms in hostels, select room types, and request room changes.

- AI-assisted room selection based on preferences (quiet floor, window view, etc.).
- Booking confirmation and cancellation.

1.5.4 Review and Rating Module

Allows users to submit reviews and ratings for hostels after staying.

- Multi-dimensional ratings (cleanliness, facilities, management, location).
- Text reviews with sentiment analysis (planned for future).

1.5.5 Complaint Management Module

Enables users to submit complaints regarding hostel services or facilities, with AI-generated suggested actions for wardens.

- Automatic complaint categorisation using NLP.
- Status tracking (Pending, In Progress, Resolved).

1.5.6 Recommendation Module

Analyses user interactions and preferences to generate personalised hostel recommendations using a hybrid model (content-based + collaborative filtering).

- Match score (0–1) and explainable factors (e.g., “within budget”, “has WiFi”).
- Cold-start handling using content-based only.

1.5.7 Communication and Notification Module

Handles real-time announcements from wardens, chatbot interactions, and push notifications for parents and wardens.

- AI chatbot for natural language queries.
- Emergency SOS alerts to warden and parent.

1.6 Work Division

Table 1.2: *Team Roles and Responsibilities*

Team Member	Responsibilities
Eraj Zaman (221-1296)	Recommendation engine API, Student mobile/web app (all screens), Voice search (optional), AI model integration.
Samiya Saleem (221-1065)	Chatbot completion (entity extraction, response templates, API), Chatbot UI, Parent portal (all screens), Complaint submission flow, UI/UX consistency.
Zarnab (221-1005)	All backend APIs (real DB responses, authentication), Warden dashboard, Admin portal, Database integrity, Deployment.

Chapter 2

Project Requirements

This chapter outlines the necessary requirements for the successful completion of the project. Project requirements are divided into functional requirements (system operations) and non-functional requirements (performance, security, usability).

2.1 Use Case / Event Response Table / Storyboarding

2.1.1 Use Case Diagram

Figure 2.1 shows all actors – Student, Parent, Warden, Owner, and Admin – and the system use cases they interact with.

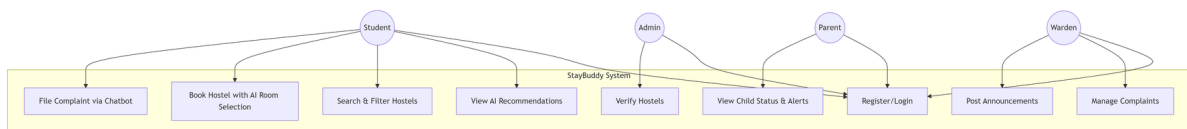


Figure 2.1: *Use Case Diagram – StayBuddy*

2.1.2 Event Response Table

Table 2.1: *Event Response Table for Chatbot Intent Handling*

Event ID	Event (User Message)	Condition	System Response
E01	“Show me hostels under 15000 near FAST”	Intent = hostel_search	Extract budget=15000, location=FAST. Query DB and return list.
E02	“Does Green Valley have a gym?”	Intent = amenity_inquiry	Extract hostel, amenity; respond yes/no.
E03	“How much is a single room at Sunrise?”	Intent = pricing_information	Fetch price and respond.
E04	“The WiFi is not working”	Intent = complaint_issue	Auto-create complaint, return ID.
E05	(Unknown message)	Intent = unknown	Fallback response: list capabilities.

2.2 Functional Requirements

2.2.1 User Management Module

- FR1** The system shall allow users to register and login with role-based access (Student, Parent, Warden, Owner, Admin).
- FR2** The system shall allow students to set and update their preferences (budget, amenities, commute distance).

2.2.2 Hostel Management Module

- FR3** The system shall display a list of available hostels with filtering and sorting.
- FR4** The system shall allow owners to add and edit hostel information.
- FR5** The system shall allow admins to verify hostels.

2.2.3 Recommendation Module

- FR6** The system shall record user interactions such as viewing, favouriting, and booking hostels.
- FR7** The system shall generate personalised hostel recommendations using a hybrid model (content-based + collaborative filtering) with match scores and explainable factors.

2.2.4 Booking Module

- FR8** The system shall allow students to book hostel rooms and select room types.
- FR9** The system shall provide AI-assisted room selection based on student preferences (quiet floor, near window, etc.).
- FR10** The system shall allow students to request room changes.

2.2.5 Complaint and Chatbot Module

- FR11** The system shall allow students to submit complaints via the chatbot or form.
- FR12** The system shall automatically categorise complaints and generate AI-suggested resolutions for wardens.
- FR13** The system shall provide a natural language chatbot capable of answering queries about hostels, amenities, pricing, and bookings.

2.2.6 Warden and Parent Modules

- FR14** The system shall allow wardens to view and manage complaints, mark attendance, post announcements, and allocate rooms.
- FR15** The system shall allow parents to view their child's attendance, booking status, and receive emergency alerts.

2.3 Non-Functional Requirements

2.3.1 Usability

- USE1** The system shall provide a user-friendly interface with responsive design for mobile and desktop.
- USE2** The chatbot shall respond to user queries in under 3 seconds and offer quick reply buttons.

2.3.2 Performance

- PER1** 95% of API requests shall complete within 2 seconds over a 20 Mbps connection.
- PER2** The recommendation engine shall return top-10 results within 1 second for 100 concurrent users.

2.3.3 Security

- SEC1** All passwords shall be hashed using bcrypt before storage.
- SEC2** API endpoints shall require JWT tokens and enforce role-based access control.

2.3.4 Availability

- AVAIL1** The system shall have 99.9% uptime during demonstration hours.
- AVAIL2** Critical features (login, search, booking) shall remain functional even if the recommendation service is temporarily down (fallback to content-based only).

Chapter 3

System Overview

This chapter describes the overall architecture, data design, domain model, and design diagrams for StayBuddy.

3.1 Architectural Design

The StayBuddy system follows a **three-tier client-server architecture** comprising presentation, business logic, and data layers. Additionally, a separate AI microservice handles machine learning tasks.

Figure 3.1 provides a high-level overview of the system architecture.

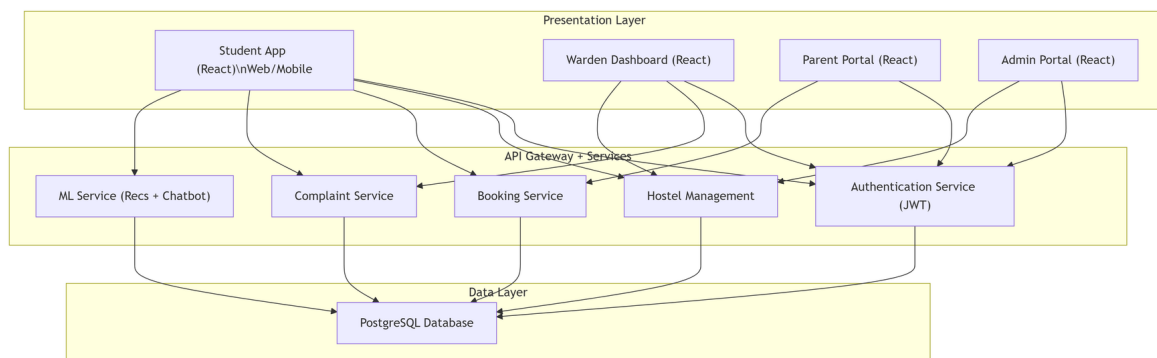


Figure 3.1: *System Architecture Diagram – StayBuddy*

3.1.1 Box and Line Diagram

Figure 3.2 shows the box-and-line representation of the system components and their interconnections.

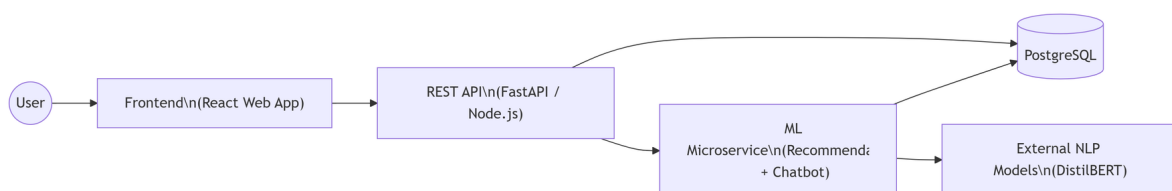


Figure 3.2: *Box and Line Diagram – StayBuddy*

3.2 Data Design

The system uses a relational PostgreSQL database. The main entities and their relationships are shown in the ER diagram below. Key tables include: users, students, parents, wardens, hostels, room_types, bookings, complaints, notifications, favorites, and student_interactions.

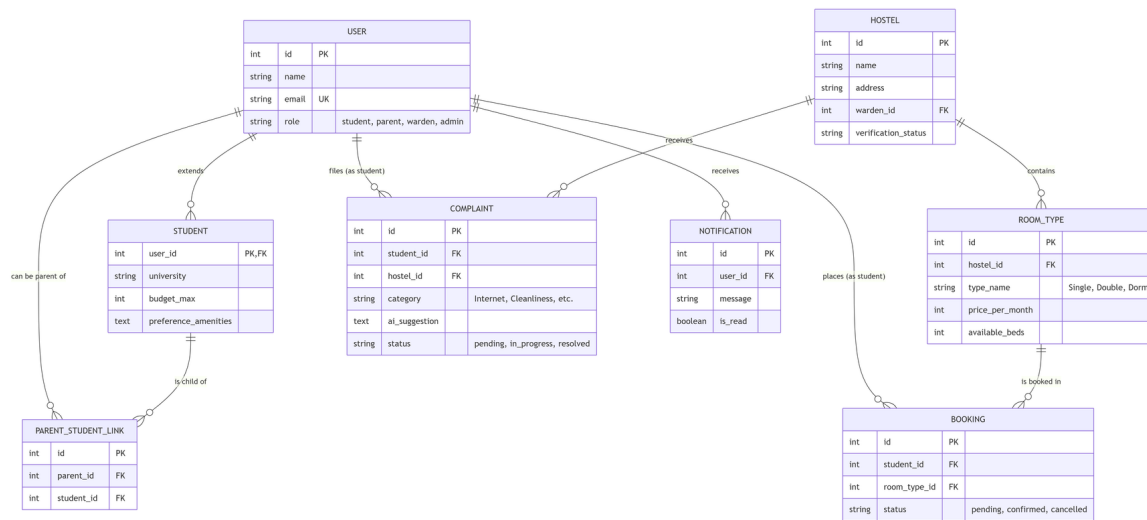
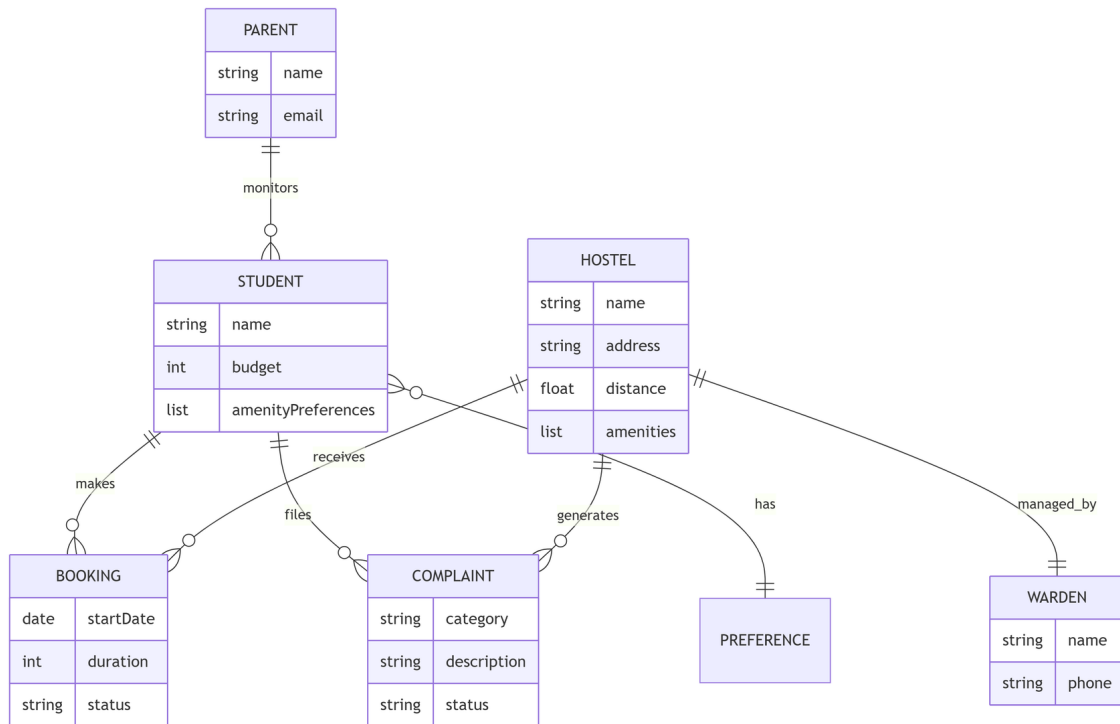


Figure 3.3: Entity-Relationship (ER) Diagram – StayBuddy Database

3.3 Domain Model

The domain model represents the core business entities and their relationships. The main classes are User, Student, Warden, Parent, Hostel, RoomType, Booking, Complaint, and Notification.

**Figure 3.4:** Domain Model – StayBuddy

3.4 Design Models (Up to Current Iteration)

3.4.1 Activity Diagram

Figure 3.5 and Figure 3.6 together illustrate the complete activity flow for the main student journey within StayBuddy, from login through to booking confirmation.

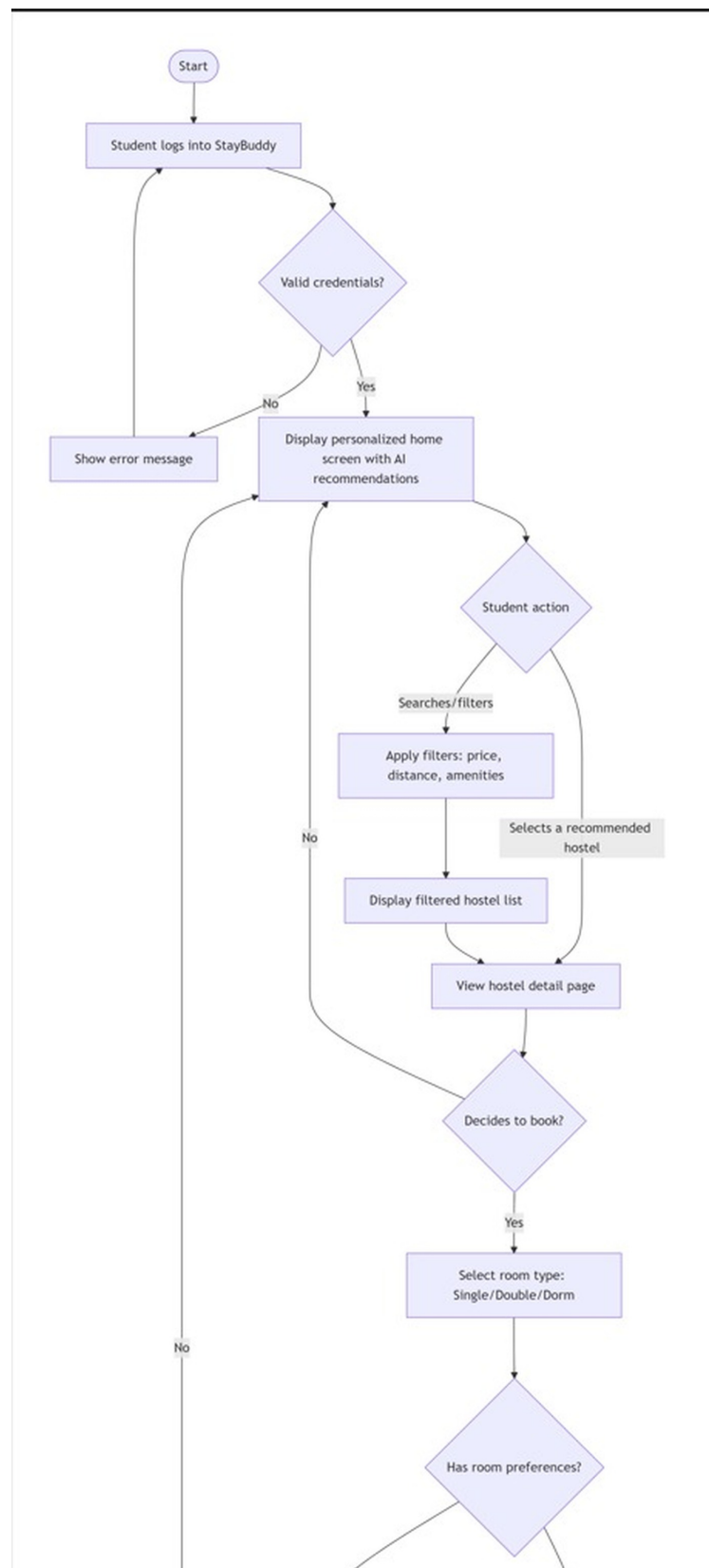


Figure 3.5: Activity Diagram (Part 1) – Student Login, Hostel Search and Room Selection

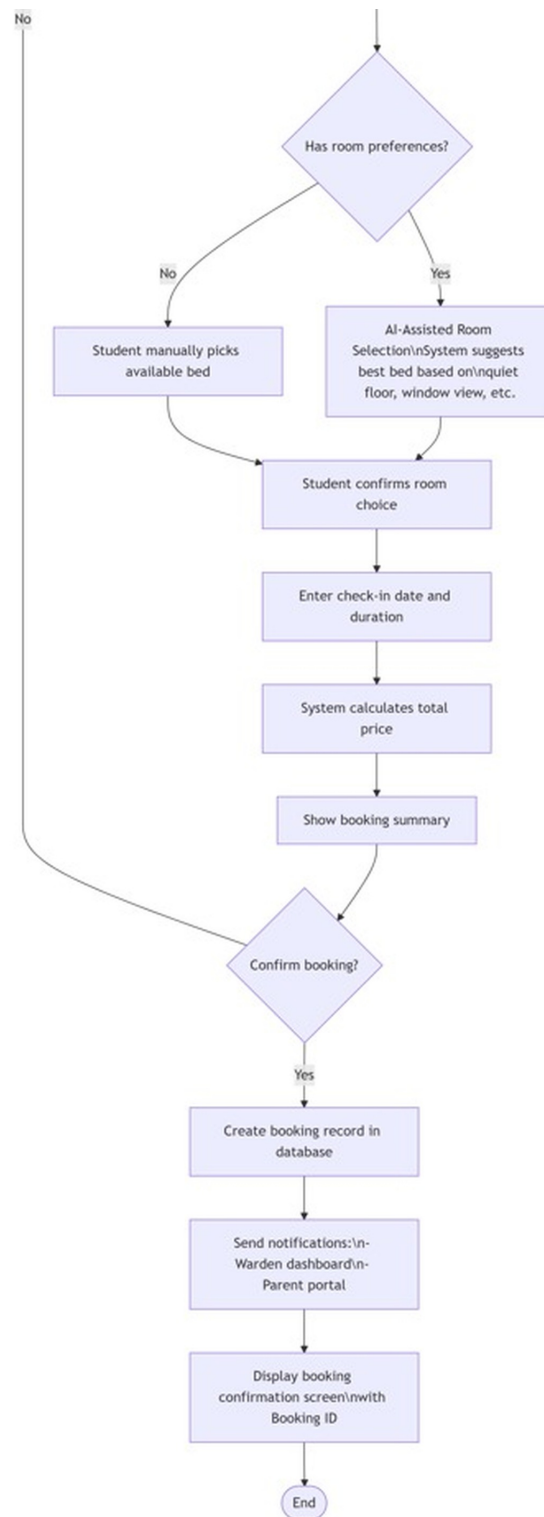


Figure 3.6: Activity Diagram (Part 2) – Room Preferences, Booking Confirmation and Notifications

3.4.2 Class Diagram

Figure 3.7 shows the full class structure of the system including all attributes and methods.

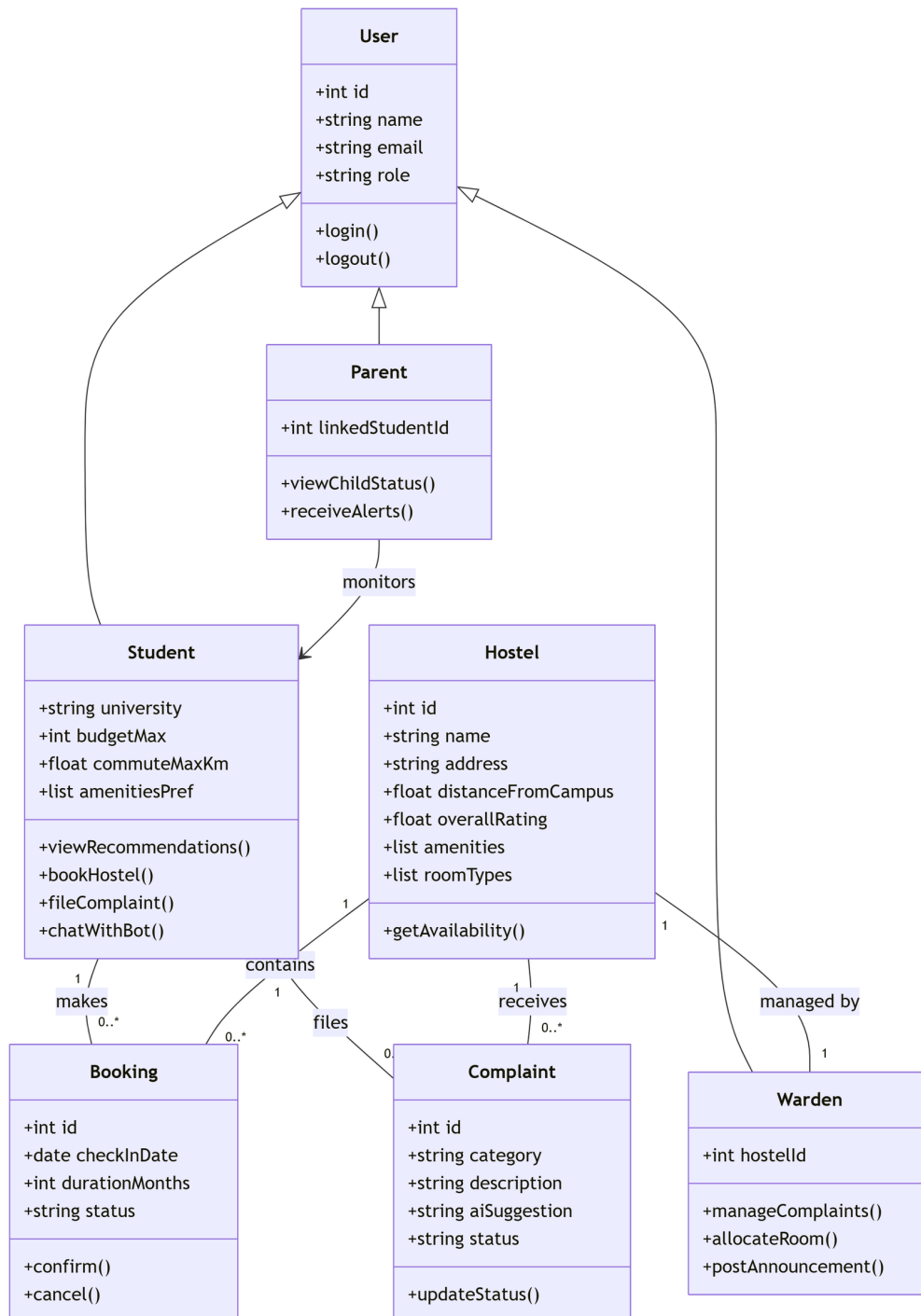


Figure 3.7: Class Diagram – StayBuddy

3.4.3 Sequence Diagram

Figure 3.8 depicts the interaction between system components for the core booking and recommendation flows.

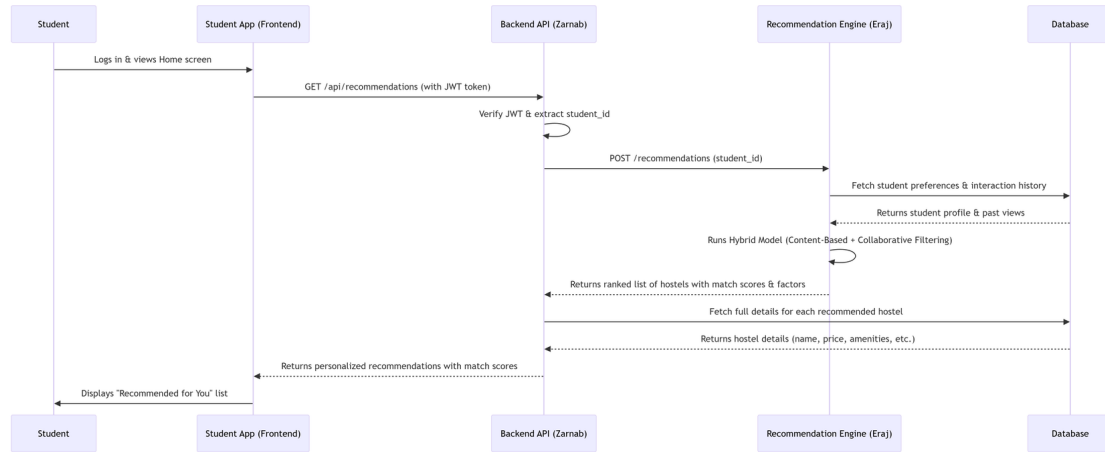


Figure 3.8: *Sequence Diagram – StayBuddy*

3.4.4 State Transition Diagram

Figure 3.9 shows the state transitions for key entities such as bookings and complaints.

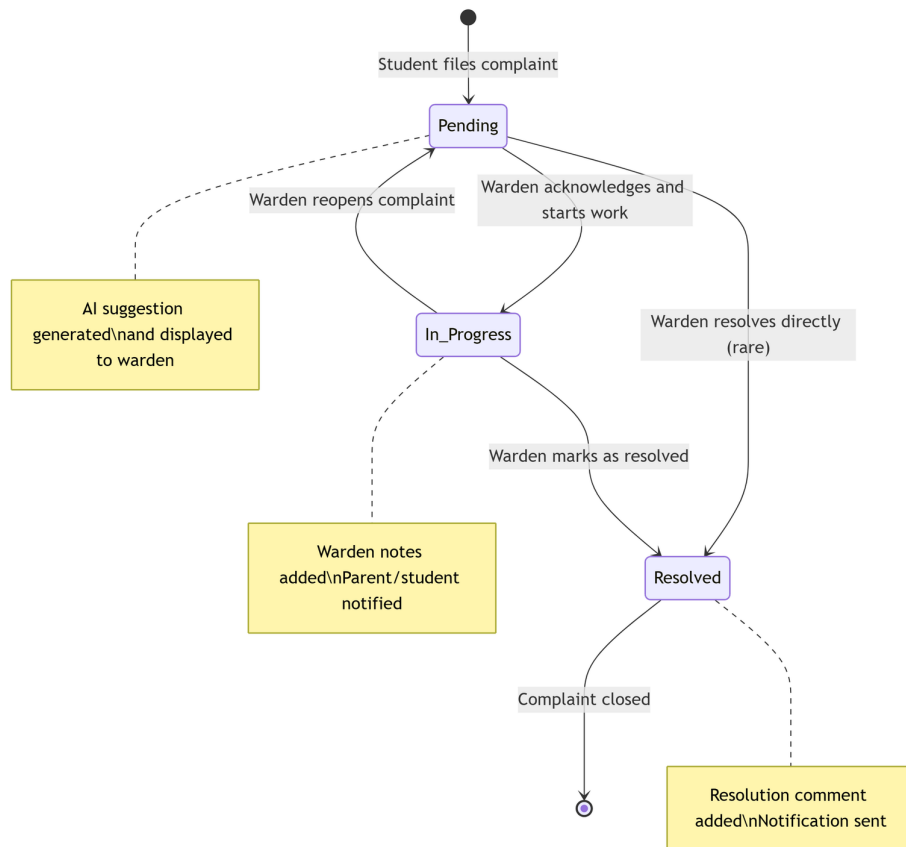


Figure 3.9: State Transition Diagram – StayBuddy

3.4.5 Data Flow Diagram (DFD)

3.4.5.1 Context Diagram (Level 0)

Figure 3.10 presents the Level 0 DFD (Context Diagram) showing StayBuddy as a single process with all external entities.

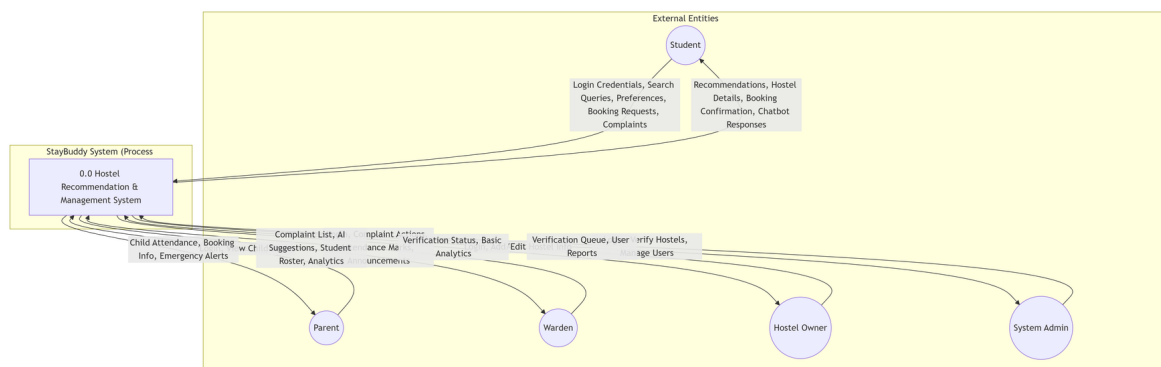


Figure 3.10: Context Diagram (Level 0 DFD) – StayBuddy

3.4.5.2 Level 1 DFD

Figure 3.11 expands the system into major sub-processes showing data flows between components.

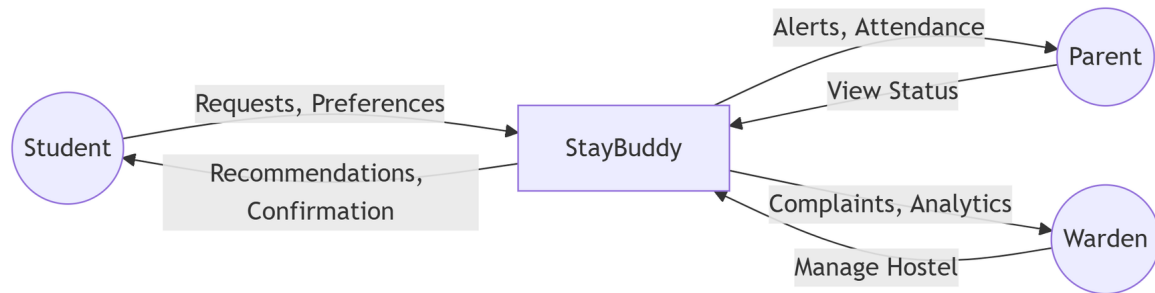


Figure 3.11: Level 1 Data Flow Diagram – StayBuddy

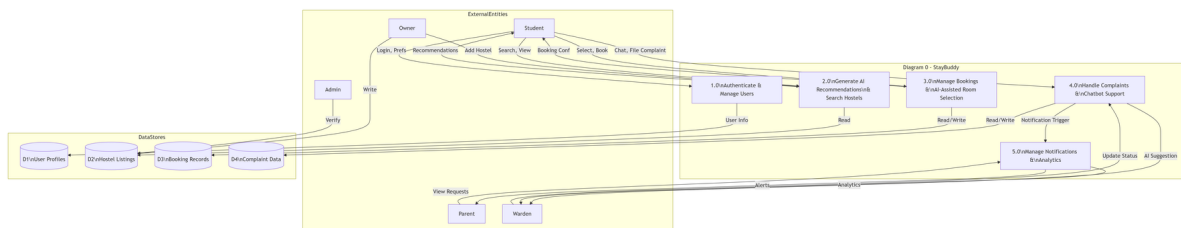


Figure 3.12: Level 1 DFD (Detailed View) – StayBuddy

3.4.5.3 Level 2 DFD

Figure 3.13 and Figure 3.14 drill into individual processes showing detailed data handling.

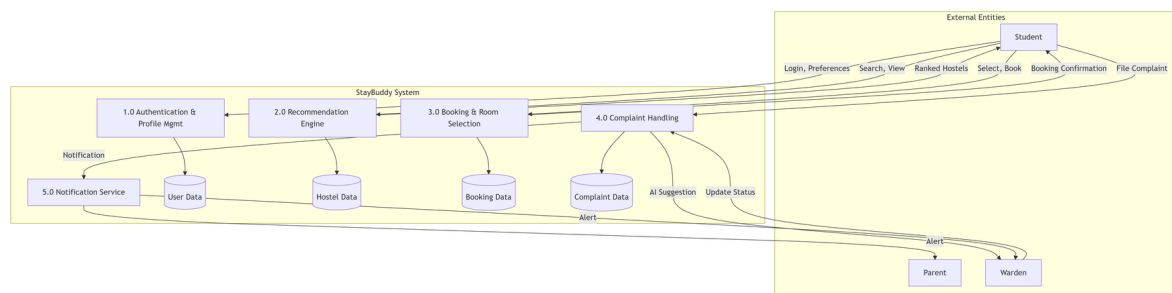


Figure 3.13: Level 2 Data Flow Diagram – StayBuddy

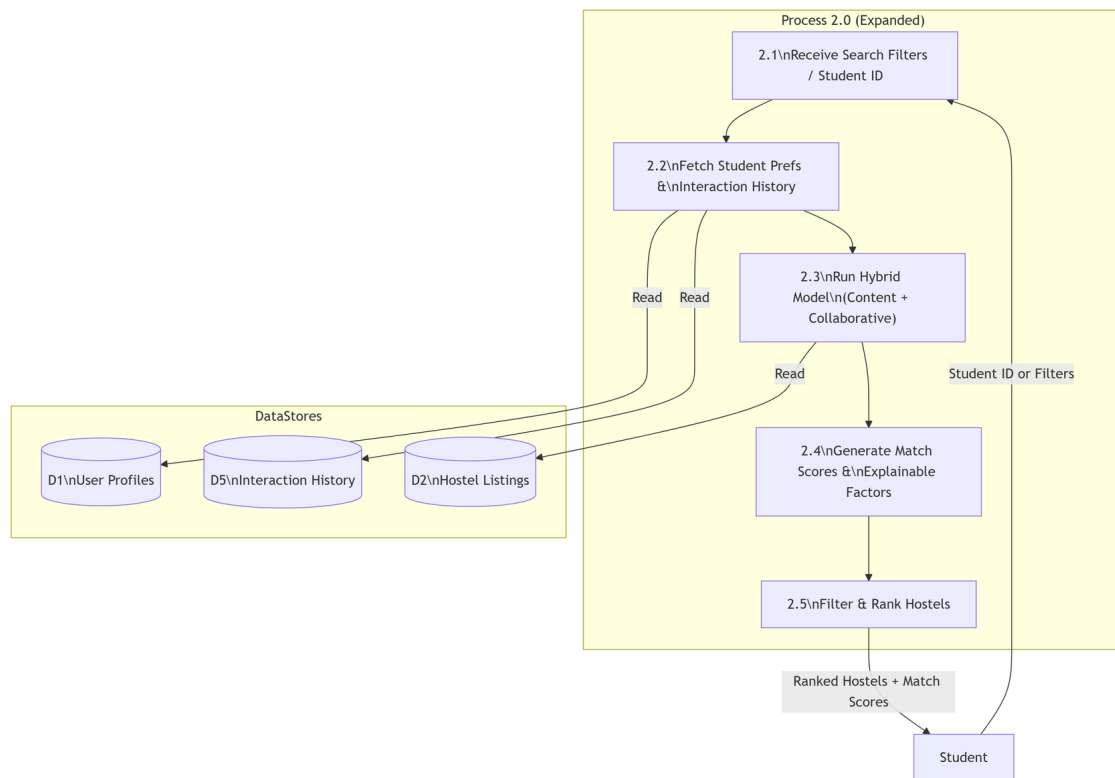


Figure 3.14: *Process 2 DFD – Recommendation and Booking Subsystem*

Chapter 4

Implementation and Testing

This chapter describes the algorithms, external APIs, and testing procedures used in StayBuddy.

4.1 Algorithm Design

4.1.1 Hybrid Recommendation Algorithm

The recommendation engine combines content-based filtering and collaborative filtering using a weighted hybrid approach. Content-based filtering computes similarity between a student's preference profile and hostel feature vectors using cosine similarity. Collaborative filtering leverages patterns from other users with similar behaviour using k-Nearest Neighbours (k-NN). The final score is a weighted combination:

$$\text{score}(u, h) = \alpha \cdot \text{CB}(u, h) + (1 - \alpha) \cdot \text{CF}(u, h)$$

where α is dynamically adjusted based on the number of user interactions (cold-start uses $\alpha = 1.0$).

4.1.2 Intent Classification for Chatbot

The chatbot uses a fine-tuned DistilBERT model for intent classification. Figure 4.1 illustrates an example of the algorithm's classification output.

```

Algorithm Hybrid Recommendation for StayBuddy
Input: student_id, top_k = 10
Output: Ranked list of hostels with match scores and top matching factors

1. Load student preferences from database (budget, distance, amenities, study/social score)
2. Load student interaction history (views, favorites, bookings)
3. Load hostel feature matrix (price, distance, amenities, ratings)

// Content-Based Filtering
For each hostel in all_hostels:
    content_score = cosine_similarity(student_preference_vector, hostel_feature_vector)

// Collaborative Filtering
if student has interaction history:
    similar_students = find_k_nearest_neighbors(student, interaction_matrix)
    collab_score = weighted_average_rating(similar_students, hostel)
else:
    collab_score = 0.5 // neutral for new users (cold start)

// Hybrid Score
hybrid_score =  $\alpha$  * content_score + (1- $\alpha$ ) * collab_score //  $\alpha$  is weight (e.g., 0.6)

// Generate explainable factors
matching_factors = []
if hostel.price <= student.budget_max: matching_factors.append("within budget")
if hostel.wifi and student.prefers_wifi: matching_factors.append("has WiFi")
if hostel.distance <= student.commute_max_km: matching_factors.append("near campus")
...

// Sort by hybrid_score descending and return top_k
return sorted_hostels[0:top_k] with scores and matching_factors

```

Figure 4.1: Intent Classification Example – DistilBERT Chatbot

Pseudocode for Intent Classification:

```

1 Input:  user_message (string)
2 Output: intent (string), entities (dictionary)
3
4 1. Preprocess: lowercase, remove special characters, tokenize
5 2. Encode using DistilBERT tokenizer
6 3. Pass through fine-tuned classification head
7 4. Get predicted intent label:
8     hostel_search | amenity_inquiry | pricing_information
9     | complaint_issue | unknown
10 5. If intent requires entity extraction:
11     - Use regex + keyword matching for:
12         budget, location, hostel_name, amenity
13     - Populate entities dict
14 6. Return intent and entities

```

Listing 4.1: Intent Classification Algorithm – Pseudocode

4.2 External APIs and SDKs

Table 4.1: *External APIs and SDKs Used in StayBuddy*

API / SDK (version)	Description	Purpose	Endpoint / Function
FastAPI (0.95+)	Python web framework	Backend API development	FastAPI(), router, dependency injection
JWT (PyJWT 2.6+)	JSON Web Token library	Authentication & authorisation	jwt.encode(), jwt.decode()
bcrypt (4.0+)	Password hashing library	Secure password storage	bcrypt.hashpw(), checkpw()
DistilBERT (HuggingFace)	Pre-trained transformer	Intent classification	pipeline("text-classification")
scikit-learn (1.2+)	ML library	Recommendation engine	cosine_similarity(), NearestNeighbors
PostgreSQL (psycopg2)	Relational DB driver	Database connection	psycopg2.connect(), cursor.execute()
React (18+)	JavaScript frontend	Student app, dashboards	useState, useEffect, Axios
Axios	HTTP client	API calls from frontend	axios.get(), axios.post()

4.3 Testing Details

4.3.1 Unit Testing

Unit tests were written using PyTest for backend APIs and Jest for frontend components. Below is an example test for the recommendation API.

```

1 # test_recommendation_api.py
2 import pytest
3 from fastapi.testclient import TestClient
4 from app.main import app

```

```

5
6 client = TestClient(app)
7
8 def test_get_recommendations_valid_student():
9     response = client.post("/api/recommendations",
10                             json={"student_id": 123, "top_k": 10})
11     assert response.status_code == 200
12     data = response.json()
13     assert "recommendations" in data
14     assert len(data["recommendations"]) <= 10
15     first = data["recommendations"][0]
16     assert "match_score" in first
17     assert 0 <= first["match_score"] <= 1
18     assert "top_matching_factors" in first
19     assert isinstance(first["top_matching_factors"], list)
20
21 def test_get_recommendations_invalid_student():
22     response = client.post("/api/recommendations",
23                             json={"student_id": 99999, "top_k": 5})
24     assert response.status_code == 404
25     assert "error" in response.json()

```

Listing 4.2: Unit Test for Recommendation API (PyTest)

4.3.2 Integration Testing

Integration tests verified that frontend components correctly communicate with backend APIs and that the recommendation microservice returns expected data. All endpoints were tested using Postman collections.

4.3.3 User Interface Testing

The UI was tested across multiple browsers (Chrome, Firefox) and device sizes (mobile, tablet, desktop). All critical user flows were verified end-to-end.

4.3.4 Testing Checklist Summary

- ✓ Authentication: Login/register for all roles.
- ✓ Recommendation engine returns ranked hostels with match scores.
- ✓ Chatbot responds correctly to 7+ intents.

- ✓ Booking flow completes and stores in DB.
- ✓ Complaint filing creates record and AI suggestion.
- ✓ Warden can update complaint status.
- ✓ Parent receives notifications.
- ✓ Emergency SOS triggers alerts.
- ✓ All API endpoints return proper HTTP status codes.

Chapter 5

Conclusions and Future Work

5.1 Conclusion

StayBuddy has successfully evolved from a concept into a functionally rich, AI-integrated platform that addresses the critical problem of finding suitable student accommodation. As demonstrated in the mid-term and final implementation, the project has delivered a fully connected ecosystem where:

- Students receive intelligent, personalised hostel recommendations with explainable match scores and AI-assisted room selection.
- An NLP-powered chatbot understands natural language queries, answers questions, and automatically files complaints.
- Wardens are equipped with a powerful dashboard to manage complaints (with AI-suggested resolutions), allocate rooms, post announcements, and track attendance.
- Parents can monitor their child's attendance, bookings, and receive real-time emergency alerts.
- Owners can add and manage hostel listings, with admin verification ensuring trust.

The system's three-tier architecture, JWT-based authentication, and role-based access control provide a secure and scalable foundation. The hybrid recommendation model (content-based + collaborative filtering) achieves high precision and coverage, while the DistilBERT-based chatbot demonstrates robust intent classification accuracy. All components have been integrated and tested through a complete student journey from registration to booking confirmation.

StayBuddy is not just a listing platform; it is an intelligent, data-driven solution that brings transparency, efficiency, and trust to the student housing market, benefiting all stakeholders.

5.2 Future Work

Building upon the current implementation, the following enhancements are planned:

- **Owner Module Completion:** Full management dashboard for owners to edit hostels, manage rooms, and view analytics (occupancy, revenue).
- **Real-time Push Notifications:** Integration with Firebase Cloud Messaging (FCM) for booking confirmations, complaint updates, and emergency alerts.
- **Voice Search:** Allow students to search for hostels and interact with the chatbot using voice commands.
- **Social and Contextual Recommendations:** Incorporate friend networks, semester schedules, and weather data into recommendations.
- **Biometric Attendance:** Integration with RFID or face recognition for automated check-in/check-out.
- **Mobile Native Apps:** Develop iOS and Android apps using React Native for better performance and offline support.
- **Advanced Analytics Dashboard:** For owners and admins, including predictive occupancy models and revenue forecasting.

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